

Causal Analysis of the USAID IQPEP in Ethiopia

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Abstract

From 2009 to 2014, USAID implemented the Improving Quality of Primary Education Program (IQPEP) in Ethiopia, aiming to bolster reading proficiency among second and third-grade students through comprehensive in-service and pre-service teacher training. While the original evaluation of this program monitored performance trends, it relied on descriptive statistics and stopped short of a formal causal inference analysis. This paper addresses that limitation by conducting a rigorous secondary analysis of the 2014 endline data to isolate the causal impact of the intervention on Early Grade Reading Assessment (EGRA) reading comprehension scores. Leveraging the randomized assignment of schools, we employ both school-level linear models and student-level methods, specifically Lin’s estimator and Causal Forests. Our results demonstrate that while school-level aggregation yields inconclusive evidence, the student-level analysis reveals a modest statistically significant positive treatment effect. Additionally, our investigation into heterogeneous treatment effects using Causal Forests indicates that these benefits were uniform across the student population, with no significant variation based on observed characteristics.

1 Introduction

From 2009 to 2014, USAID launched the Improving Quality of Primary Education Program (IQPEP), across eight different regions of Ethiopia. The stated goals were to improve reading proficiency in second and third grade students by offering training programs to teachers in service and pre-service. Many forms of evaluation were taken, including interviews, surveys, focus group discussions, and classroom observations. We focused on the Early Grade Reading Assessment (EGRA) reading comprehension score to quantitatively measure the main goal of improving reading proficiency. The EGRA tests students in phonemic awareness, decoding, fluency, vocabulary and reading comprehension. 80,347 teachers and 10,112 principals were trained through this program, in addition to many administrative officers. A multi-stage stratified sampling procedure was used to select 240 schools from the eight regions, then the selection of the 20 students from each grade was done by stratifying by sex to ensure balance among participants.

Data was collected at three points. In 2010 at the start of the program, 2013 as a midline, and finally in 2014 at the end of the program. The report noted a limitation where the evaluation team began their final evaluations in 2014 on June 13, when classes were already over or in their final examinations, and thus may have had an impact on scores. The original paper found that within the intervention schools, the number of students reaching the benchmark level of EGRA between 2010 and 2013 increased from 3% to 5% for second graders and from 11% to 16% for third graders. Interestingly, these numbers dropped between 2013 and 2014 to 3.2% for second graders and 11.5% for third graders.

Table 1: EGRA Results Summary

Assessment period	Grade Level	Planned Target	Achievement		
			Moderate	Benchmark+	Combined
2010 Baseline	Grade 2	3.1	24	3	27
	Grade 3	11.3	35	11	46
2013 Midline	Grade 2	20	28	5	33
	Grade 3	30	38	16	54
2014 End line	Grade 2	35	23	3.2	26.2
	Grade 3	50	38	11.5	49.5

The numerical analysis of the original paper stops there before doing any causal analysis with the data, and rather focusing on survey data from the participants. They do not give a numerical value to the effectiveness of the program, so this was a great opportunity to use the causal inference tools we used in class. We had a clear treatment, whether or not the school was an IQPEP-participating school or not, as well as a clear outcome, the EGRA reading comprehension score. With the treatment and outcome of interest established, we will attempt to find the effect that the IQPEP program had on reading comprehension scores of second and third graders in Ethiopia.

2 Related Works

A similar analysis was done by Araya, Rolleston et al. where they measured the impact of Ethiopia's General Education Quality Improvement Programme Phase II (GEQIP-II), focusing on math scores in grade 4 students measured at 2012 and 2019 from 33 public primary schools across 6 regions. Since they had access to many student-level covariates, they used propensity score matching by student, while achieving covariate balance at the school level. They leveraged the fact that they had data over time, matched a student to a past student, and found the causal effect as if they were the same student. Their study found that scores actually decreased a significant amount, especially in rural areas. However, they argued that the program may have been making progress crediting a "value added" statistic, where the treated students displayed a greater improvement throughout the year than the control students, despite having on average lower scores. While some of this can be attributed to the fact that the students in 2019 had more room to learn to begin with, it is also important to note that enrollment in treated schools rapidly increased from 2012 to 2019 because of the program, which could have led to the lower average scores for the treated group. Overall, their results indicated that the GEQIP-II program may have had a positive impact on student math scores but the authors were not able to establish a causal link between the intervention and an increase in absolute scores. The present study seeks to establish this causal link by utilizing a more robust methodology.

3 The Data

Our dataset contained 9406 students from May of 2014 with information on their region, grade, gender, age, correct letter per minute, correct familiar words per minute, correct invented words per minute, oral reading fluency, total correct reading comprehension questions, and many other similar speed, accuracy, and comprehension metrics. It also includes binary indicator for treatment, where the treatment is whether or not the teachers at the student's school received the training program. The organization did not publish the 2013 "midline" data referenced in the study, so we are only using the "endline" dataset.

In terms of data preprocessing, we were fortunate to have a relatively clean dataset with very few outliers. The only problematic data was the age of the students, with about 1% of students missing age data. Looking at the other covariates, there was no correlation between the students with missing age data. There was also one outlier where a second grader was listed as being 102 years old. We simply removed these observations as they were an insignificant proportion of our data.

We will now set some notation that will be used throughout the remainder of the paper. The outcome will be denoted by Y , the covariate matrix will be denoted by X , and the treatment vector will be denoted by Z . Our data is at the student level. Hence, $Y \in \mathbb{R}^N$, $X \in \mathbb{R}^{N \times P}$, and $Z \in \mathbb{R}^N$ where Y_i is the outcome of student i , X_{ij} is the j^{th} covariate of student i , and $Z_i \in \{0, 1\}$ is the treatment indicator for student i . We have 9406 students in our dataset, each with 5 covariates (age, sex, grade, language, school), a treatment assignment, and an outcome (reading comprehension score). This gives $N = 9406$ and $P = 5$. For part of our analysis, we will consider school-level effects, so we must also have notation for aggregated data at the school level. There are $K = 240$ total schools in our data, 120 assigned control and 120 assigned treatment. For each school, we compute the average of the outcomes, which we will denote by $Y^{(k)} \in \mathbb{R}^K$ where $Y_i^{(k)}$ is the average outcome for the i^{th} school. Because treatment was assigned at the school level, we can denote $Z^{(k)} \in \mathbb{R}^K$ where $Z_i^{(k)} \in \{0, 1\}$ is the treatment indicator for the i^{th} school. We must similarly aggregate our covariates at the school level to get the following values at the school level: number of 2nd graders, number of 3rd graders, number of females, number of males, average student age, standard deviation of student age, language. The number of covariates has increased to 7, so we denote $P^{(k)} = 7$. We denote the covariate matrix to be $X^{(k)} \in \mathbb{R}^{K \times P^{(k)}}$ where

$X_{ij}^{(k)}$ is the j^{th} (aggregated) covariate for the i^{th} school.

As previously stated, we will get estimates of causal effect at both the school level and the student level. The outcome is the reading comprehension score of a student, taking values in $\{0, 1, 2, 3, 4, 5\}$. We aggregate reading comprehension scores at the school level to get an estimate of the causal effect. The distribution for scores in treatment and control schools is shown here. Note, the two distributions show here are $Y^{(k)} \cdot I(Z^{(k)} = 0)$ (red) and $Y^{(k)} \cdot I(Z^{(k)} = 1)$ (blue).

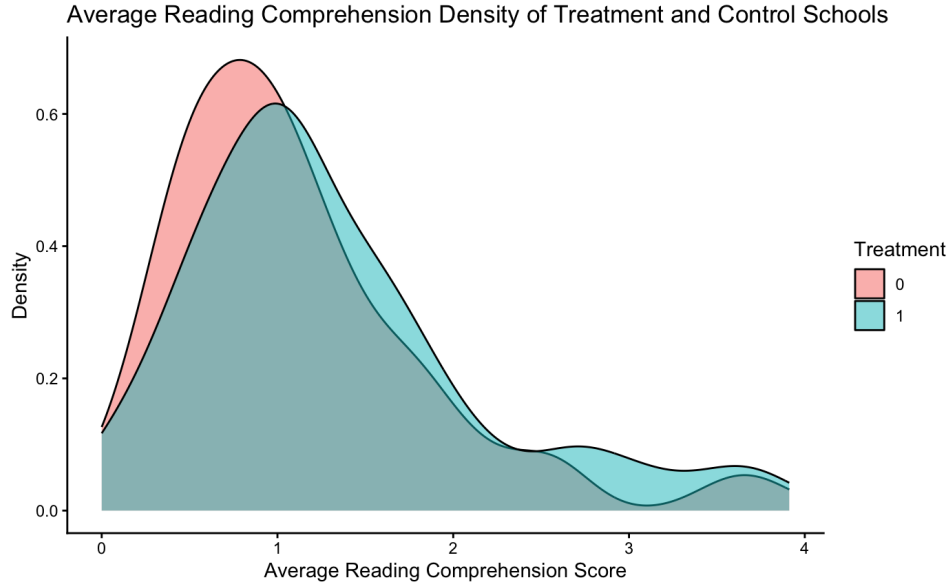


Figure 1: Average school outcomes by treatment assignment.

We see that both distributions are right-tailed, but the control distribution has center further to the left, indicating that the students at those schools performed worse, on average, in the reading comprehension examination. Again, this is a good sign of the efficacy of the treatment, which we will later examine to see if it can be formalized into a causal claim.

Similarly, the scores for each student by treatment group is shown here.

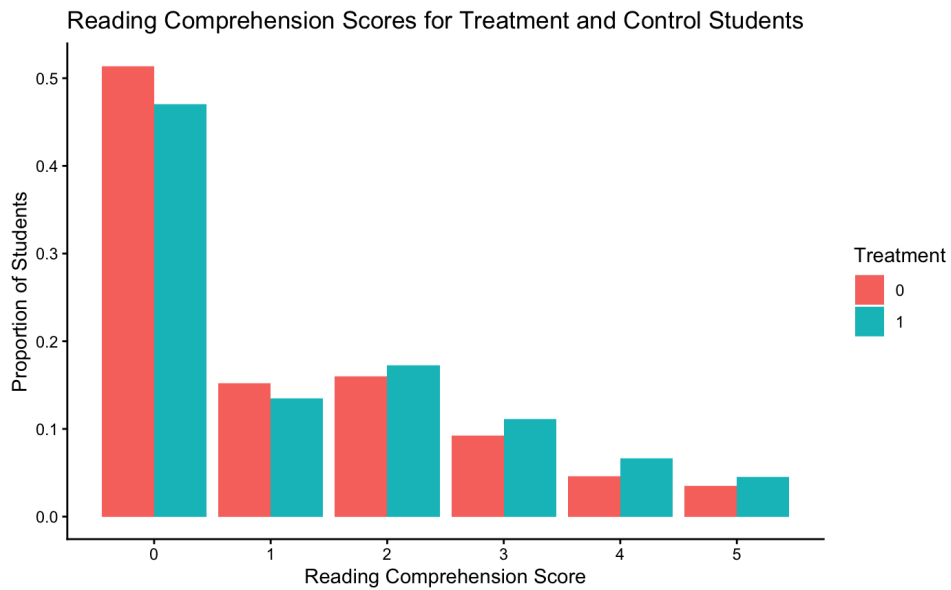


Figure 2: Student outcomes by treatment assignment.

It appears that students who received the treatment have a higher proportion of scores in $\{2, 3, 4, 5\}$, and students who didn't receive the treatment have a higher proportion of scores in $\{0, 1\}$. This is a good sign, and we will attempt to make causal claims about this relationship when we conduct our student-level analysis.

4 Methods and Analysis

4.1 School-Level Analysis

In the experimental setup, treatment was randomly assigned at the school level, allowing our school-level analysis to be relatively straightforward. Our research question for school-level analysis will be this: Does the treatment lead to better average reading comprehension by students in the treated school? Because we have a completely randomized experiment, we will ignore our covariates and use a linear model to estimate the average treatment effect (ATE), $\tau = E[Y^{(k)}(1) - Y^{(k)}(0)]$. Again, $Y^{(k)}$ is the vector of average reading comprehension scores at each school.

Because of the aforementioned randomization, we should not have any difference in covariate distribution between students in treated schools and students in control schools. As a quick sanity check, we will make sure this is the case. To do this, we examine the standardized difference in covariates between treatment and control schools, shown here.

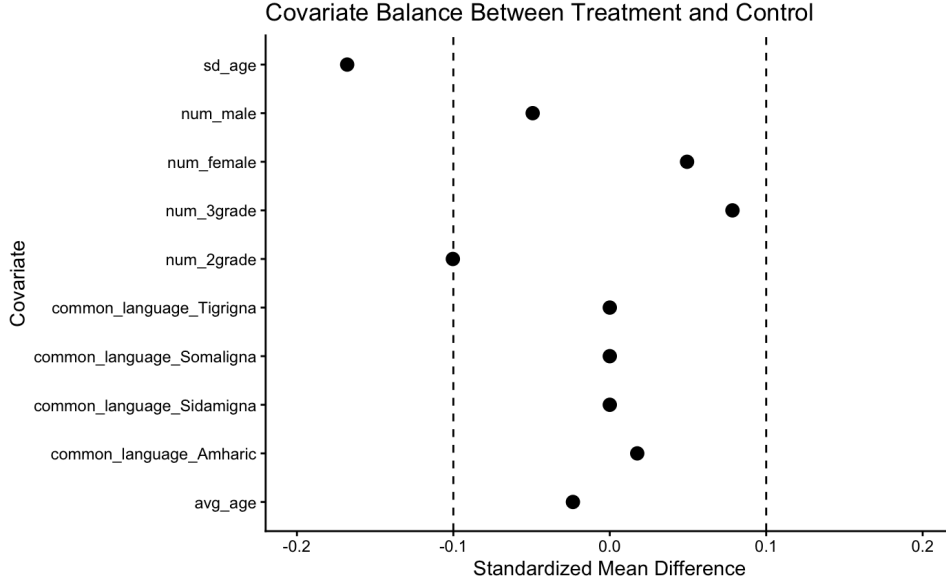


Figure 3: Covariate imbalance at the school level.

We can see that all of the standardized differences are within $[-0.1, 0.1]$ except for the standard deviation of age, indicating that control schools have a marginally higher spread in the age of their students. This difference is still small and it is only one covariate, so we will continue this section without considering our covariates.

We will also make SUTVA assumptions. In this case, this means that the outcome for one school is not affected by the outcome for other schools and all students at treated schools received consistent treatment. We do not have any data that allows us to empirically validate these assumptions, but we can note that the distance between many of the schools makes interaction unlikely and the training program given to teachers at treated schools was consistent.

We also assume the ATE can be effectively approximated by a linear model given by:

$$Y_i^{(k)} = \alpha + \hat{\tau} Z_i^{(k)}$$

Using this model, we get the following results: ATE estimate $\hat{\tau} = 0.190$, standard error estimate $\hat{SE}(\hat{\tau}) = 0.103$, and p-value $\hat{p} = 0.0663$. These results also give the 95% confidence interval $[-0.012, 0.392]$. This confidence interval containing 0, along with our p-value being greater than 0.05 leads us to conclude that there is not a significant causal effect of the treatment at the school level. Our estimate $\hat{\tau} = 0.190$ implies that treated schools, on average, have an average reading comprehension score that is 0.190 higher than control schools.

While the setup of the experiment greatly simplified this approach, it does certainly have limitations. The main limitation that we noted is that it fails to account for individual student factors. When looking at the distribution of reading comprehension scores above, we see a right skew, which indicates that most students performed average on the reading comprehension exam, but there were a few students who performed exceedingly well. It would be beneficial to attempt to single out these students through student-level covariates as they have a disproportionately large weight on our school-level estimates given that we are taking school-wide averages.

4.2 Student-Level Analysis

We will now shift our focus to the student level, which will allow us to incorporate covariates that may directly impact student performance on the reading comprehension exam. We suspect that this may lead to improved results that are not susceptible to the skew of the reading comprehension distribution. In addition, this will allow us to utilize variance reduction techniques that might render the results statistically significant. Our research questions for student-level analysis will be following: (1) Does the school-level treatment lead to better reading comprehension scores for individual students on average? (2) Is the treatment more effective for students with certain characteristics or learning specific languages? These questions give rise to the following estimands:

$$\begin{aligned} ATE : \tau &= E[Y(1) - Y(0)] \\ CATE : \tau(X) &= E[Y(1) - Y(0)|X] \end{aligned}$$

The treatment being randomly assigned at the school level presents an interesting challenge. We must somehow find a way to "group" students in the same school together as they will experience a great level of interaction. If unaccounted for, this scenario will lead to correlation within the standard errors of our estimate. This leads us to the solution of cluster-robust standard error calculations. The standard error calculations in this section will employ CR0 clustered sandwich estimators, which allows arbitrary correlation of errors within clusters while assuming independence across clusters. The specific formula for variance of our least squares estimator is:

$$\hat{V} = X^T X \left[\sum_{k=1}^K (X^{(k)})^T (\hat{\tau}^{(k)})^T \hat{\tau}^{(k)} X^{(k)} \right] X^T X$$

where $\hat{\tau}^{(k)}$ is the vector of least square residuals within cluster k .

4.2.1 Analysis with Lin's estimator

In order to analyze the data at the student level we utilized Lin's Estimator (Lin, 2013) with school clusters to account for randomization at the school level. Lin's estimator is an ordinary least squares (OLS) regression specification with robust standard errors that adjusts for baseline covariates by including them not only as main effects, but also as interaction terms with the treatment indicator. By using centered covariates (subtracting the mean from each feature), this fully interacted model ensures that the coefficient on the treatment variable provides a consistent and asymptotically efficient estimate of the ATE, even if the underlying linear model is misspecified. Under this framework, the outcomes are given by:

$$Y_i = \alpha + \hat{\tau} Z_i + X_{ci}^\top \hat{\beta} + (X_{ci} \cdot Z_i)^\top \hat{\gamma} + \epsilon_i$$

Where $\hat{\tau}$ is the estimated ATE and X_{ci} are the centered covariates for unit i . In addition to achieving

an efficient variance reduction for our estimate, this estimator also allows us to measure the conditional average treatment effect (CATE), and thus heterogeneous treatment effects, given by:

$$\hat{\tau}(x) = \hat{\tau} + X_{ci}^\top \hat{\gamma}$$

Table 2: Lin’s estimator results at the student-level

Variable	Estimate	Std. Error	t value	p-value	95% CI
(Intercept)	1.112	0.060	18.630	0.000	[0.99, 1.23]
z	0.189	0.092	2.052	0.041*	[0.01, 0.37]
z:grade	0.086	0.084	1.021	0.308	[-0.08, 0.25]
z:female	-0.007	0.066	-0.112	0.911	[-0.14, 0.12]
z:age	-0.025	0.025	-0.983	0.327	[-0.07, 0.02]
z:language_Amharic	-0.109	0.225	-0.485	0.628	[-0.56, 0.34]
z:language_Sidamigna	-0.355	0.188	-1.886	0.063	[-0.73, 0.02]
z:language_Somaligna	-0.298	0.514	-0.580	0.564	[-1.33, 0.73]
z:language_Tigrigna	-0.276	0.194	-1.423	0.160	[-0.66, 0.11]

The results of the regression using Lin’s estimator are displayed in Table 2. The estimated ATE was 0.189, which was statistically significant ($p = 0.041$). However, taking into account the range of the reading comprehension scores of 0 through 5, this suggests a very modest improvement in the outcome. This is further demonstrated by the lower bound of the 95% CI being very close to zero. Finally, no significant CATEs were detected as the p-value of all interaction terms was above 0.05. However, this last result could be driven by non-linear relationships between covariates and treatment, which this model would not be able to detect.

4.2.2 Analysis with Causal Forests

To further explore heterogeneous treatment effects at the student level, we utilized Causal Forests (Wager & Athey, 2018), maintaining clustering at the school-level. While Lin’s estimator relies on global linear interactions and covariate centering to achieve robustness, Causal Forest employs ”orthogonalization”. This process locally residualizes the outcomes and treatment assignment to isolate the causal effect, ensuring consistency without imposing a global parametric form. Under this framework, the rigid linear specification is replaced by:

$$Y_i = \hat{\mu}(X_i) + \hat{\tau}(X_i)Z_i + \epsilon_i$$

Where $\hat{\mu}(X_i)$ represents the baseline prognostic function ($E[Y|X, Z = 0]$), which is predicted by the random forest model, and $\hat{\tau}(X_i)$ is the estimated CATE. This distinction addresses the primary limitation of Lin’s estimator: the assumption of linearity. Instead of restricting the heterogeneous effects to a fixed linear slope ($X_{ci}^\top \hat{\gamma}$), the Causal Forest learns $\hat{\tau}(X_i)$ non-parametrically. This allows the model to adapt to the data locally, capturing complex, non-linear heterogeneities that the standard interaction model would fundamentally miss.

Finally, $\hat{\tau}$ (ATE) is calculated utilizing the forest’s predictions as an input to the Augmented Inverse Probability Weighting (AIPW) estimator, given by:

$$\hat{\tau} = \frac{1}{n} \sum_{i=1}^n \left(\underbrace{\hat{\tau}(X_i)}_{\text{Forest Prediction}} + \underbrace{\frac{Z_i - \hat{e}(X_i)}{\hat{e}(X_i)(1 - \hat{e}(X_i))} (Y_i - \hat{m}(X_i) - (Z_i - \hat{e}(X_i))\hat{\tau}(X_i))}_{\text{Correction Term}} \right)$$

Where $\hat{e}(X_i)$ is the estimated propensity score ($E[Z = 1|X]$) and $\hat{m}(X_i)$ is the estimated conditional mean outcome ($E[Y|X]$). It is worth noting that given the randomized nature of the data, the benefits of utilizing AIPW will be marginal given that the covariates are already balanced between treatment and control.

In our analysis we did not perform manual cross fitting or parameter tuning as the algorithm employs automated tuning and "honest" estimation. Specifically, to prevent overfitting, each tree utilizes data splitting, where one subsample determines the split structure, while a separate subsample estimates the effects within the leaves. This separation decouples model selection from estimation, which is theoretically required to guarantee asymptotically normal estimates and reliable confidence intervals (Wager & Athey, 2018).

Table 3: Causal Forests results at student-level

Variable	Estimate	Std. Error	p-value	95% CI
ATE	0.190	0.091	0.036*	[0.01, 0.37]
MFP	1.063	0.265	0.000*	-
DFP	-173.225	7.517	1.000	-

The results of the causal forest model are displayed in Table 3, which are in line with those from the analysis utilizing Lin’s estimator. The ATE is estimated at 0.190 and is statistically significant ($p = 0.036$). The Mean Forest Prediction (MFP) is close to 1 (1.063) and highly significant ($p < 0.001$), confirming that the forest successfully captured the average treatment signal. However, the Differential Forest Prediction (DFP) yields a p-value of 1.000. This lack of statistical significance indicates that the model failed to detect valid heterogeneity, providing no evidence that treatment effects vary systematically across students.

The inclusion of a limited set of covariates has important implications for the analysis. The low dimensionality constrains the model’s ability to detect heterogeneous treatment effects. The lack of significant heterogeneity detected by the model indicates that the treatment effect does not vary systematically across the observed dimensions (Grade, Age, Gender, Language). However, this does not rule out the possibility that heterogeneity exists along unobserved dimensions not included in the model.

5 Conclusion

At the school level, our analysis was straightforward given the completely randomized experiment with treatment assigned at the school level. We conclude that schools that received the training, on average, had a 0.190 higher average reading comprehension score, which was found to be statistically insignificant after an examination of standard errors. At the student level using Lin’s estimator, we found a very similar causal effect. We conclude that students who attended school that received the training, on average, had a 0.189 higher reading comprehension score. This result was determined to be statistically significant because we were able to reduce variance due to the usage of student-level covariates. In our experimentation with causal forests, we also found that students who attended schools that received the training, on average, had a 0.190 higher reading comprehension score. We were able to achieve an even lower standard error with this method and concluded that our result was statistically significant. However, we note that the model did not detect any heterogeneity in our data.

Despite the over 90,000 educators trained through the program, the original report noted that personnel turnover was an issue throughout all parts of the education system. This means that there is an issue of the sustainability of the training and thus the results as we get further from the start of the program.

Between 2010 and 2014, the proportion of students attending primary school increased from 71.82% to 80.87% (UNESCO, 2021). This changes the population of the students in both treatment and control schools, which adds another complication to finding the true causal effect of the program.

In Araya, Rolleston, et al. they noted that the largest impact that GEQIP-II had was on rural students, but our data was incomplete in that the urban/rural column in our dataset was blank, and there was no way to fill that information in with the school codes being encoded. With the way IQPEP structured the schools from resource centers to satellite schools, it is plausible that having the urban/rural status would have helped.

In further research, if we had any reports or estimates about the turnover rate of teachers, it may

be interesting to look into models that are able to accommodate non-compliance. Another approach would be to use the approach that Araya, Rolleston, et al. used since we had data of second and third graders from 2010 before the implementation of IQPEP, so we could have done propensity score matching between the students in 2014 and 2010, and use that to find a similar average treatment effect on the treated (ATT). Finally, conducting this analysis with a higher number of covariates might uncover heterogeneous treatment effects that were missed by the causal forest analysis, given the low-dimensional nature of our data.

6 Works Cited

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7 Code Repository Link

All of our finalized code for this project can be found in the file named FinalCode.Rmd in this GitHub repository: <https://github.com/trentbellinger/Stats209-FinalProject>.